

# MEHDI BELHAJ

## Power Electronics & Hardware Engineer

Analog & digital electronics · Embedded systems · Power conversion — Aerospace · Space · Defence · Energy

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270 VDC

SiC 1200 / 1700 V

10 kW

DO-160G · DAL A

EMC / EMI

OrCAD · KiCad

Matlab · PSIM · LTSpice

C · C++ · Python

|              |                     |                            |                                |             |                   |
|--------------|---------------------|----------------------------|--------------------------------|-------------|-------------------|
| AVAILABILITY | EXPERIENCE          | MOBILITY                   | STATUS                         | LICENCE     | ENGLISH           |
| Immediate    | 3 years · aerospace | Toulouse · France · Europe | French work permit → Nov. 2028 | B — own car | C1 · professional |

### PROFILE

**3 years of experience** designing high-voltage electronic boards for the aerospace industry (Airbus, Safran, SuperGrid Institute): Flyback, Buck, SEPIC and LDO power supplies · three-phase SiC 1200 V inverter — 10 kW · DO-160G EMC filtering · analog and digital sensing chains. **Full V-cycle ownership**, from specification to qualification testing — author of the key deliverables (design justification file, PCB routing specification, test plans) and active across the industrial process (Windchill PLM, part codification, procurement). C/C++ & Python — Cisco / OpenEDG certificate tracks (330 h).

### TECHNICAL SKILLS

|                             |                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| POWER ELECTRONICS           | Three-phase SiC 1200 / 1700 V inverter — 10 kW · Flyback 540 V · Buck · SEPIC · LDO · Active DC rectifier · Isolated gate drivers UCC21750 (DESAT) · RCD snubbers · DC bus pre-charge · Inrush current limiting · Magnetics design (transformers, inductors)                                                                                               |
| ANALOG & SENSING            | Current / voltage sensors (ACS37002, INA190, shunts) · ADC MAX1363, ADS1247 24-bit · Temperature TMP1075 · Force · Position · Op-amps · Signal conditioning & filtering · Galvanic isolation                                                                                                                                                               |
| DIGITAL & EMBEDDED          | TI DSP TMS320F280049 (C2000) · PIC16F · Embedded C · C++ (OOP, templates, STL) · Python · VHDL (basics) · UART · SPI · I2C · RS422 · PWM · ADC · DAL A software (equipment SW, sequencer, BITE) · Git · Linux · LabVIEW · PyTest                                                                                                                           |
| EMC · PROTECTIONS · TESTING | DO-160G Section 21 — cat. B & H · Conducted & radiated emissions (LISN) · Common-mode / differential-mode filters · Lightning pin injection · 15 kV ESD · Dielectric tests up to 20 kV · Insulation measurements · IEC 60068 · Bench hardware debugging, oscilloscope                                                                                      |
| CAD & PCB DESIGN            | OrCAD Capture CIS (schematics, BOM) · KiCad · Proteus · IPC-2221 routing specification · HV clearances with altitude correction (IEC 60664-1) · 6-layer PCB · Stack-up · DFX & testability · Mechanical integration on 3D models                                                                                                                           |
| SIMULATION & CONTROL        | Matlab / Simulink / Simscape · Simulink Control Design · Scilab / Xcos · PSIM · LTSpice · Saber · Small-signal stability · PI controller synthesis · Bode · Nyquist · Nichols · FFT                                                                                                                                                                        |
| PROCESS & INDUSTRIALIZATION | Full V-cycle · Internal design & gate reviews · Technical requirement specifications · Hardware verification procedures · Design justification file · Compliance matrix · Windchill PLM · Part codification & BOM · Supplier consultations · Obsolescence & second sourcing · ABD0100 requirements · DO-254 (basics) · Product compliance CE / RoHS / WEEE |

### PROFESSIONAL EXPERIENCE

SAFRAN

**Power Electronics & Hardware Engineer — Safran Ventilation Systems**  
Via Centum T&S · Toulouse (Blagnac), France · Electronics lead · High-voltage DC networks · More Electric Aircraft (MEA)  
Nov. 2023 — Jan. 2025 · 14 mos

**Design of an embedded power supply and energy conversion solution for ventilation systems on the 270 VDC network — DO-160 category H. Complete Safran process, V-cycle covered autonomously.**

- **Hardware design — conversion & power supplies:** input stage board (EMI filter + isolated Flyback, LDO and Buck supplies); 9 functions designed and justified: reverse-current protection, phase loss < 4 ms, ride-through of interruptions ≥ 200 ms, inrush limiting, lightning, 15 kV ESD, over-temperature; 1,890 W, -40...+100 °C.
- **PCB design & routing:** author of the PCB routing specification — 6-layer FR4 PCB, stack-up, IPC-2221 clearances corrected for altitude (IEC 60664-1), DC bus tracks 9 A RMS, DFX & testability rules; **mechanical integration:** interface drawing, thermal exchange zones, connector placement, iterations with the mechanical design office on the 3D model.
- **Aerospace qualification:** DO-160G cat. H — common-mode / differential-mode EMC filters, RCD snubbers, PSIM + FFT simulations; writing and execution of EMI test plans (LISN, conducted & radiated emissions), lightning and insulation tests.
- **Industrial documentation — V-cycle:** author of the design justification file (per-function justifications, calculation notes), requirement specification, hardware verification procedure, compliance matrix, capitalization sheet; facilitation of multidisciplinary meetings (HW, layout, mechanical, procurement), internal design & gate reviews; weekly interface in English with Safran USA.
- **Configuration management, PLM & procurement:** component codification and bills of materials (BOM) in Windchill; supplier consultations and selection, part numbering, aerospace qualification (temperature, altitude, derating), obsolescence & second sourcing.

**Results:** functional mock-up EMI-validated, complete design package delivered, 270 VDC feasibility demonstrated with minimal software and mechanical impact.

AIRBUS

**Power Electronics & Hardware Engineer — Airbus / CROUZET**  
Via Serma Ingénierie · Toulouse, France · R&D zero-emission programme · DAL A criticality · ABD0100 requirements · EMC DO-160G Section 21 cat. B  
Dec. 2022 — Nov. 2023 · 11 mos

**PoC of an ultra-fast electromechanical actuator (EMA) deploying a spoiler in ≤ 130 ms — Serma · CROUZET · Airbus partnership: 3 boards (CPU, HVDC ±270 V, Inverter) within 65×65×170 mm.**

- **Requirements, specifications & Safety:** contribution to the definition of hardware & software requirements and specifications, and to the Safety study with Serma Safety; compliance with ABD0100 aerospace requirements.
- **SiC gate driver board:** power supply, biasing and gate driver design — isolated UCC21750 driver (DESAT) for 6 SiC 1200 V MOSFETs; three-phase inverter 10 kW pulsed; quasi-resonant Flyback 540 V→24 V without optocoupler; Buck 24→12 V (brake / locking).

- **Analogs & sensors:** high-voltage current and voltage sensing chains (MAX1363 ADC), INA190 current sensing, TMP1075 temperature, force-measurement chain — 24-bit ADC ADS1247; conditioning and filtering.
  - **Digital & software:** CPU board with TMS320F280049 DSP (UART, SPI, I2C, RS422, PWM, ADC); contribution to the design and coding of the DAL A embedded software (equipment software, sequencer, BITE); writing of the sub-assembly specifications.
  - **Motor control — Matlab / Simulink / Simscape:** system simulation, controller design, parameter identification;  $\leq 130$  ms dynamics validated (Bode / Nyquist); LTSpice & PSIM simulations.
  - **Reliability assessment:** SiC junction thermal analysis in pulsed operation, ageing risks, reliability reviews with the partners.
- Results:** 130 ms requirement validated in simulation; hardware architecture approved in Airbus & CROUZET design reviews — PoC integrated into the zero-emission studies.



### Design Studies, Modelling & Control Engineer — Safran Tech

Via AKKODIS (formerly Aeroconseil AKKA) · Toulouse, France · DC voltage regulation & power sharing — active rectifier

Aug. — Dec. 2022 · 5 mos

- **Research & studies:** power-stage modelling using a 2<sup>nd</sup>-order transfer function; PI controllers for DC bus voltage regulation and current sharing between parallel converters; identification of the dynamic parameters (damping  $M_v$ , bandwidth  $\omega_v$ , bus capacitance  $C_{bus}$ ).
- **Testing & validation:** simulation of the current loops, field weakening and load-step responses; stability and robustness through Bode / Nyquist analyses — PM  $46^\circ \rightarrow 61^\circ$ , GM  $21 \rightarrow 24.8$  dB,  $\sim 265$  ms response; controller tuning recommendations.
- **Full modelling training:** MATLAB / Simulink, Simscape, Scilab / Xcos, Control System Toolbox, Simulink Control Design.

**Results:** controllers validated with compliant margins (GM  $> 6$  dB, PM  $> 45^\circ$ ); robustness demonstrated on constant-power loads; linearization methodology and tuning recommendations delivered.



### Graduate Internship — R&D Power Electronics Engineer — SuperGrid Institute

Villeurbanne (Lyon), France · Research laboratory · HVDC grids · Isolated HV supply for series-connected SiC gate drivers

Feb. — Jul. 2022 · 6 mos

- **Research & development:** isolated HV power supply with 12 cascaded ferrite transformers driving 6 series-connected SiC 1.7 kV MOSFETs (high / low side); selection of TDK EPCOS T38 cores.
- **Laboratory work & testing:** prototyping, oscilloscope measurements, dielectric tests up to 20 kV, analysis of common-mode currents under  $dV/dt \leq 16$  kV/ $\mu$ s.
- **Studies & industrialization:** LTSpice simulations correlated with measurements; specification for industrial prototyping (efficiency, EMC).

**Results:** 20 kV insulation validated, 12-transformer cascade functional, industrial specifications delivered.

## ENTREPRENEURSHIP — IN PARALLEL

### Co-founding partner — TOKYOSEAT — side venture, run in parallel

Electrical & sanitary equipment · French SAS with 5 partners · TOKYOSEAT INPI-registered trademark · Pan-European e-commerce distribution — Amazon FBA & Shopify (FR, DE, IT, ES, NL) and UK · long-term investment

Since Jul. 2023

- **Product & regulatory compliance:** CE marking (Low Voltage, EMC), RoHS, UKCA — laboratory testing, technical files and EU declarations of conformity, multi-country regulatory watch; INPI trademark registration (classes 9 & 11).
- **Environmental compliance:** WEEE & packaging EPR registrations for France / Germany (Stiftung EAR register); German quality label “SEHR GUT (1.3)” earned for the flagship product (DasQualitätssiegel.de independent testing, Oct. 2025).
- **Sourcing, supply chain & management:** factory selection and monitoring, sampling campaigns, product improvement driven by customer-feedback analysis; import (EORI, TARIC classification, customs, VAT), freight forwarders; business plans, margin & ROI calculations, cash-flow dashboards.

**2025 – 2026:** German and UK market development (quality label, WEEE compliance, UKCA marking), run alongside the intensive C++ / Python training below. Fully outsourced logistics (Amazon FBA · Shopify with a third-party fulfilment centre) and day-to-day operations run by a salaried manager: venture stabilized, compatible with a full-time engineering role.

## EDUCATION



**2020 — 2022 · National 5-year degree (Bac+5) · Master's grade**

**Master's degree in EEA — Electronics, Electrical Energy, Automatic Control · engineer-level** — track Electrical Energy: Conversion, Materials, Sustainable Development · specialization in Power Electronics, Actuators & Control — Université Toulouse III Paul Sabatier · INP Toulouse – ENSEEIHT

## CONTINUING EDUCATION & CERTIFICATIONS



### Continuing education — C++ & Python software development — Cisco / OpenEDG

In partnership with OpenEDG C++ Institute & Python Institute · certificate tracks aligned with the CPE / CPA / CPP and PCEP / PCAP professional frameworks · 5 verifiable Credly badges

2025 — 2026 · 330 h

- **C++ Advanced (CPP) & C++ Essentials 1 & 2 (CPE / CPA):** function and class templates, full STL (sequential & associative containers, sorting, merging and set algorithms, iterators), stream & file I/O, full OOP, pointers & memory management, exceptions, namespaces — the foundation of real-time embedded C++ and safety-critical software.
- **Python Essentials 1 & 2 (PCEP / PCAP):** OOP, iterators & generators, modules / packages / PIP, file handling — applied to test automation and measurement post-processing (test benches, EMI, oscilloscope).
- **Published projects** — [github.com/mehdi-belhaj-eea](https://github.com/mehdi-belhaj-eea): scope-emi-toolkit (EMI pre-check, 8 tests) · embedded-dsp-cpp (C++17/STL, 12 tests) — green CI.

**Badges:**



**2024 · French electrical safety authorization H0 · B2V · B2V Essai · BE Essai** — CAMIRA training body: work and testing on low- and high-voltage installations



**2024 · EYE Berlin — European Youth Event · European Parliament · EYP & Schwarzkopf Foundation** — 1,500 young Europeans, Berlin 18-20 April 2024: workshops, panels and debates on the European elections, innovation and sustainability

## LANGUAGES

French — **bilingual** · English — **C1**, fluent (weekly Safran USA meetings) · Spanish · Portuguese — basics

## SOFT SKILLS & INTERESTS

Autonomous · rigorous · team player · proactive · Competitive football · NQT mentoring · travel — European tour, 18 countries visited